4.11.5 Wrap-Up: Reflection

1. Complete the learning pyramid.

One question I would like to have answered …	
One mistake I learned from today …	One thing I am not sure about … YET!
One thing I am confident I know is …	

Unit 4, Lesson 12: Writing Equations of Parallel and Perpendicular Lines

Benchmark

MA.912.AR.2.3 Write a linear two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.

Learning Targets

- ☐ I can write an equation for a line parallel to a given line through a given point.
- ☐ I can write an equation for a line perpendicular to a given line through a given point.

4.12.1 Warm-Up: Bell Work

1. Write the equation of the line that goes through the point $(-27, 10)$ and is parallel to the line $y + 5 = \frac{5}{9}(x - 10)$.

2. Write the equation of the line that goes through the point $(-27, 10)$ and is perpendicular to the line $y + 5 = \frac{5}{9}(x - 10)$.

4.12.2 Collaboration: Sorting Equations

1. Work with your partner to sort the equations into pairs that have the given characteristic. Find three different sets of pairs of equations for each characteristic. Lines may be used more than once.

Pairs of Parallel Lines	**Pairs of Perpendicular Lines**	**Pairs of Lines with the Same y-intercept**

A	$5x + 4y = 40$	**G**	$y + 25 = -\frac{5}{6}(x - 60)$
B	$y = 6x + 1$	**H**	$3x - y = {}^{-}1$
C	$y - 11 = 3(x - 2)$	**I**	$y = -\frac{5}{6}x + 24$
D	$-3x + 2y = -12$	**J**	$y = \frac{5}{4}(x - 16)$
E	$y = -\frac{5}{4}x - 2$	**K**	$x + 6y = {}^{-}12$
F	$4x + 5y = 120$	**L**	$y = -\frac{2}{3}x + 5$

2. Work with your partner to determine which pair of lines have the same x-intercept.

   A. A and G
   B. B and J
   C. D and K
   D. F and G

3. Discuss with your partner which key features will be the same for any pair of parallel lines.

   ☐ decreasing/increasing
   ☐ domain
   ☐ range
   ☐ slope
   ☐ x-intercept
   ☐ y-intercept

4. Discuss with your partner which key features will be the same for any pair of perpendicular lines.

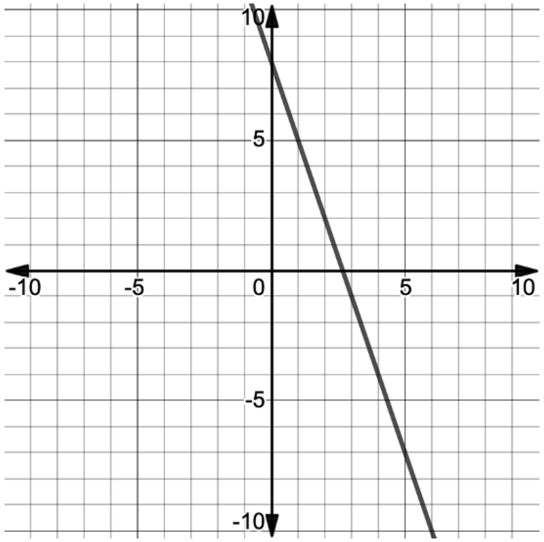
   ☐ decreasing/increasing
   ☐ domain
   ☐ range
   ☐ slope
   ☐ x-intercept
   ☐ y-intercept

4.12.3 Your Turn!

1. Write the equation of the line, in all three forms, that is perpendicular to the line $6x - 3y = 15$ and passes through the point $(1, -5)$.

Point-Slope Form	Slope-Intercept Form	Standard Form

2. Write the equation of the line, in all three forms, that is parallel to the line shown on the coordinate grid and that passes through the point $(-4,-7)$.



Point-Slope Form	Slope-Intercept Form	Standard Form

3. Write the equation of the line that passes through $(5,9)$ and is parallel to the line that passes through the points $(-4,-1)$ and $(-1,-1)$.

4. Write the equation of the line that passes through $(-12,15)$ and is perpendicular to the line that passes through the points $(-2,-4)$ and $(3,6)$.

4.12.4 Wrap-Up: Error Analysis

1. The work of two students is shown in the table. The students were asked to write the equation of the line that passes through $(-5,-9)$ and is perpendicular to the line that passes through the points $(3,-4)$ and $(0,5)$. Find each person's error and correct it.

Sandip	Jiang
Slope $= \frac{change\ in\ y}{change\ in\ x} = \frac{-4-5}{3-0} = -3$	Slope $= \frac{change\ in\ y}{change\ in\ x} = \frac{-4-5}{0-3} = 3$
Equation: $y+9=-3(x+5)$	Equation: $y+9=-\frac{1}{3}(x+5)$

2. What is the equation of the line that passes through $(-5,-9)$ and is perpendicular to the line that passes through the points $(3,-4)$ and $(0,5)$?

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